



Impacts of different characteristics of marine biofouling on ship resistance

Youjia Zou^a, Xixi Zhou^{a,b}, Lixiong Chen^a, Xiangying Xi^{c,*}

^a Merchant Marine College, Shanghai Maritime University, China

^b School of Engineering and Applied Science, University of Pennsylvania, PA, USA

^c Wuhan University of Technology, China

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ABSTRACT

Ship resistance research on marine biofouling is an old but hot topic which invokes plenty of studies recently. Previous studies on the surface roughness of ship hulls mainly focus on the height of micro-organisms or macro-organisms adhered to ship hulls without considering other components which may indeed play an important role in generating additional resistance. Many authors attempt to use a single parameter, for example, height of the biofoulers, to describe the hull roughness. Our research indicate that any single factor affecting ship hull roughness can not completely reflect the compositions of ship resistance. Here we propose a novel numerical model which enables more precise estimates of the contribution of each single component involving in the marine biofouling.

The standard $k-\epsilon$ model of FLUENT is modified by considering the possible rotating, jetting, surging and swirling of fluid flowing over ship hulls due to characteristics of the marine biofoulers. Simulated results using the modified model at various speeds are compared with that using the original model, and an improvement of ship resistance is evident, quantitatively consistent with measured, indicating that the modified model behaves in a reasonable manner in simulating the relationship between ship resistance and ship surface roughness.

The marine biofoulers affect ship resistance primarily by increasing the tangential surface shear stresses. Simulated results demonstrate that additional resistance caused by height, shape, hardness and density of the biofoulers accounts for up to ~80% of total resistance in the heavy fouling condition (taking vessel “YOURIXX” as an example), implying an economic demand for good hull performance.

1. Introduction

In 2018, the bulk carrier “YOURIXX”, with full laden of iron ore 180,000 metric tons, loaded in Tubarao of Brazil, was heading for Tokyo Bay. Unfortunately, the return trip seemed not so smooth as usual with an unprecedented reduction of average speed by about 3 knots before arriving at Singapore. Sea condition from south Atlantic to south Indian Ocean, including Cape of Good Hope, was generally peaceful. No anomalous adverse currents were experienced, either, during entire voyage. Chief Engineer and his assistants examined main engine, rudder, propeller, etc., and nothing was found in a malfunction condition, except that load of the main engine obviously increased by about 30% and daily fuel oil consumption hit a ceiling (by about 10%). More seriously, loss of days superimposed on significant fuel oil penalties resulted in a heavy claim from the charterer.

The unhappy experience of a 3-month waiting for cargo at the anchorage of Tubarao is likely reminiscent of a possible hull biofouling

due to marine bacterial adhesion based on previous practice. Therefore, a thorough submarine scan and examination were arranged by ship-owner no sooner than the ship arrived at Singapore anchorage.

Post underwater examinations reinforced the suggestion that 30–40% of hull in water was colonized by barnacles, mussels, oysters, etc., creating extra resistance. It was thought to be the most severe category among the three categories of calcareous biofouling in the shipping industry (Class NK, 2019).

After a 2-day quick removal of the calcareous biofouling at Singapore anchorage using submarine robots, the ship resumed its normal speed.

The marine species adhesion can occur on many immersed structures in seawater (Leroy, 2006), such as vessels, oil platforms, mooring buoys, etc. The hulls of merchant ships are the ideal substrate of the marine aquatic species to adhere and proliferate (Yuan et al., 2017). The disadvantages of the biofouling for all commercial ships are evident. It not only reduces ship performances, increases hydrodynamic resistance, causes corrosive damage to ship hulls (Schultz et al., 2015), but also increases ship shaft power penalties and emits more carbon dioxide and

* Corresponding author.

E-mail address: xxy98982018@163.com (X. Xi).

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Nomenclature

C_b	block coefficient
C_f	frictional resistance coefficient
C_{fm}	form resistance coefficient
C_t	total resistance coefficient
C_w	residuary resistance coefficient
C_v	viscous resistance coefficient
F_{rs}	ship's Froude number
R_{am}	total resistance after modified
R_{bm}	total resistance before modified
R_T	total resistance
ΔR_T	increment of total resistance
ΔR_f	increment of total frictional resistance
UDF	user defining function

other pollutants into air, deteriorating the atmospheric environment (Flemming, 2002), seriously against the ideology of “Green Navigation”. Furthermore, the marine biofoulers on ship hulls can lead to a spread of non-native invasive species (Drake and Lodge, 2007), threatening global ecological environment. Presently, mitigation of the marine species adhesion is becoming a high priority for some shipping companies.

Generally, the marine biofouling can be classified into 3 categories (Yao et al., 2017): the first is the mild biofouling induced by micro-foulers in a relatively short idle period. The marine micro-organisms adhered to ship hulls can be washed away when ships move forward at a normal speed. The second is the moderate biofouling mainly caused by microfoulers plus some macrofoulers, with the marine organisms sticking to shell plates being difficult to be washed out as ships move forward at their own normal speeds, but an extra forward speed may work better. The increased resistance in both category 1 and 2 is mainly frictional resistance, with other resistances increased, such as viscous resistance, form resistance, wavemaking resistance (a.k.a residuary resistance), etc., playing a minor role. The third category, however, is the heavy biofouling primarily induced by macrofoulers in a long idle period. The marine organisms on ship hulls are not possible to be washed out by ships' own forwarding in water. More aggressive methods, such as scrapping, sandblasting or sonicating, against these attachments, are required (Ceri et al., 1999). The increased resistance involves not only the frictional resistance but also the viscous resistance, form resistance and wavemaking resistance.

Previous studies on the relationship between ship resistance and surface roughness mainly focus on the frictional resistance and wavemaking resistance based on the equations implemented in the FLUENT software through computational fluid dynamics (CFD) but other added resistances, such as the form resistance, viscous resistance, etc., remain inadequate analyzing (e.g., Demirel et al., 2014; Schultz, 2007). Our investigations show that the viscous resistance arising from slimes secreted by the marine species is not the same as that caused by seawater viscosity (Xiong and Yuan, 2016). The surface tangential shear stresses produced by the former, but neglected by many authors in previous literature, are much stronger than previously thought. The results from other research show that slimes secreted by some biofoulers can result in a significant growth of resistance (Schultz, 2007), qualitatively consistent with our suggestion. The form resistance generated by the biofoulers (e.g., the macro-organisms) also play a non-negligible role in reducing ship speeds and increasing power penalties. A severe calcareous biofouling (e.g., in Fig. 1c) is able to result in ~86% penalties of propulsive powering in maintaining a normal service speed (Schultz, 2007).

Underestimates even negligence of other important resistance are bound to lead to sizable errors in estimating the total drag of ships. In essence, the surface of a ship hull fouled by different marine organisms,

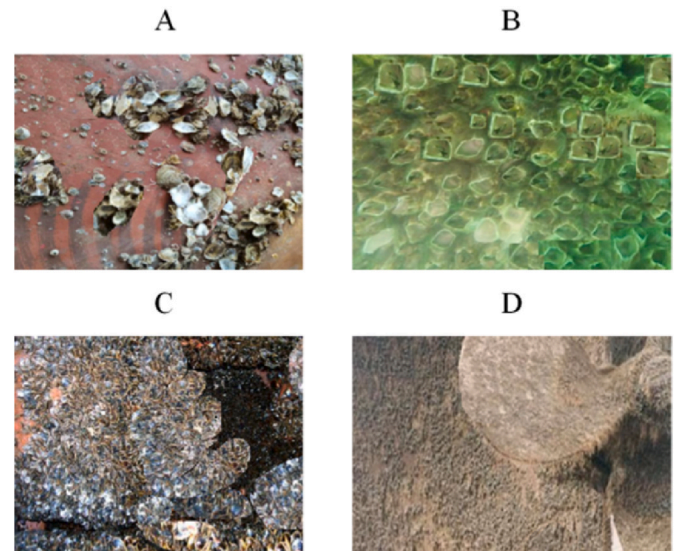


Fig. 1. Biofouling conditions. a, shells on the hull. b, underwater scanning. c, barnacles on ship hull. d, micro-organisms on ship stern part. Figs c and d are from <https://baike.baidu.com/item/%E9%99%84%E7%9D%80%E7%89%A9/4312375?fr=aladdin>.

such as barnacles, mussels and oysters, is quite different from a rough surface defined by textbooks. In this case, the hull surface roughness is far beyond the definition of traditional roughness, with extra height and sharp form (Fig. 1). In addition, different shapes (protuberances) of the marine species on ship hulls can generate different form resistances. Occasionally, this discrepancy of the form resistances can be huge. For example, sharp bulges (e.g., barnacles and oysters) on ship hulls tend to largely modify turbulent boundary layers but round bulges (e.g., tube-worms) usually do not. Therefore, whether the previous equations and coefficients are still applicable or not is a great concern, challenging our understanding.

Novel methods for resolving above issues are largely hampered by knowledge gaps. Here we develop a suite of equations based on the theory of the computational fluid dynamics after fully considering the types, shapes and distributions (coverage) of the marine organisms. The results obtained from the simulations are validated against the well-established archive records, demonstrating that the designed model behaves in a reasonable manner in simulating the ship speeds with additional resistances arising from extra roughness of the hull surface.

This paper is dedicated to an improvement of the current model in simulating the relationship between ship speeds and extra roughness of the ship hulls by developing a suite of new equations which can better characterize the hydrodynamic performance of ships in the context of unusual hull roughness caused by marine species (rather than a simple numerical simulation). Here a novel philosophy which is different from previous studies on the biofouling is proposed by comprehensively considering shape, height, hardness and density of the marine organisms because any single factor (e.g., height of the marine organisms) which is widely cited and used by previous authors (e.g., Demirel et al., 2014; Song et al., 2019) can not perfectly reflect the roughness created by these macro-organisms. The remaining of this paper are generally arranged as follows: section 2 provides materials and methods in constructing configurations of the modified models; section 3 presents the simulated results and compared them with that obtained with the original model; the discussion is presented in section 4 and the last section is the conclusion.

2. Materials and methods

Estimates of the total ship drag can be yielded through numerical

methods and towing tank experiments as well. In this study, a modified numerical method is developed based on the CFD.

2.1. Turbulent boundary layer and coordinate system

One can assume that, as a ship moves through the water, the ship is thought to be fixed, but the flow runs towards the ship. The origin of the coordinate system is fixed on the ship, creating a no-slip condition. The innermost layer of the turbulent boundary layer ($y = 0$) is “still” relative to the ship, with velocity u in any direction (x, y, z) identically vanishing (Larsson and Raven, 2010b). The outermost layer of the turbulent boundary layer, however, travels at a same speed (u_0) as the ship ($u_{ship} = u_0$) along x direction, forming a velocity gradient between the innermost and outermost layers of the turbulent boundary layer. u_x grows with y moving away from the ship hull within the turbulent boundary layer (Larsson and Raven, 2010a). Thickness of the turbulent boundary layer is determined by type, shape, height, strength, density of biofoulers and viscosity of fluid (Fig. 2).

2.2. Seawater flowing over a rough hull

Due to turbulent boundary layers existing around a ship hull, the type, shape, height, strength, density of the biofoulers and viscosity of fluid will ultimately deplete some extra propulsive power through increasing turbulent kinematic energy and enhancing surface tangential shear stresses.

The frictional resistance, viscous resistance and form resistance are largely modified owing to the existence of the biofoulers on the hulls but the wavemaking resistance remains little change because the forces acting on the bow due to pressure change very little for most merchant vessels with speed $< 20\text{--}25$ kts (Yao et al., 2017). Therefore, we focus on the growth of the frictional resistance, viscous resistance and form resistance.

2.3. Boundary conditions

Appropriate boundary conditions are necessary to be selected prior to simulations because these conditions directly influence the precise flow solutions. The calculation area of ship surface flow field is a hexahedron, including air and water. Its boundary includes inlet boundary, outlet boundary, hull, side boundary of computational domain and upper and lower boundary (including top and bottom), as shown in Fig. 3.

Velocity inlet is set to be the inlet boundary condition while pressure outlet is set to be the outlet boundary condition (assuming $p = \rho gh$, p is pressure, ρ is water density, g is gravitational acceleration and h is water depth).

The ship hull has a no-slip wall boundary condition. Since the hull has two opposite faces at the x -direction of the domain, therefore, a symmetry plane is utilized to halve the required cell numbers and to

model a quarter of the hull. This can result in smaller numbers of cells and less computational time without significantly lowering the computational accuracy.

The locations of the boundaries are shown in Fig. 4. The other boundary-related conditions are chosen to be similar to those given in (Chen and Yang, 2012).

2.4. Model equations

The time-averaged continuity equation is given in equation (1) (Chen and Yang, 2012):

$$\frac{\partial p}{\partial t} + \frac{\partial}{\partial x_i} (\rho u_i) = 0 \quad (1)$$

Reynolds time-averaged momentum equations shows in equation (2):

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial}{\partial x_i} (\rho u_i u_j) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial u_i}{\partial x_j} - \rho \overline{u'_i u'_j} \right) + s_i \quad (2)$$

where.

- ρ is density of fluid,
- μ is dynamic viscous coefficient,
- u_i and u_j are time-averaged components of velocity,
- u'_i and u'_j are pulse components of velocity,
- p is time-averaged pressure,
- s_i is the generalized term of momentum equation,
- “ $\cdot$ ” is time-averaged value.

In the widely-used standard $k - \epsilon$ model of FLUENT, k is turbulent kinetic energy and ϵ is turbulent dissipation rate:

$$\epsilon = \frac{\mu}{\rho} \left[\frac{\partial u'_i}{\partial x_k} \right] \left[\frac{\partial u'_i}{\partial x_k} \right] \quad (3)$$

Turbulent viscosity μ_i can be expressed as a function of k and ϵ :

$$\mu_i = \rho C_\mu \frac{k^2}{\epsilon} \quad (4)$$

In this model, k and ϵ are unknown variables, and their transport equations can be described as equations (5) and (6), respectively:

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k u_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_i}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k + G_b - \rho \epsilon - Y_M + S_k \quad (5)$$

$$\frac{\partial(\rho \epsilon)}{\partial t} + \frac{\partial(\rho \epsilon u_i)}{\partial x_i} = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_i}{\sigma_\epsilon} \right) \frac{\partial \epsilon}{\partial x_j} \right] + C_{1\epsilon} \frac{\epsilon}{k} (G_k + C_{3\epsilon} G_b) - C_{2\epsilon} \rho \frac{\epsilon^2}{k} + S_\epsilon \quad (6)$$

where.

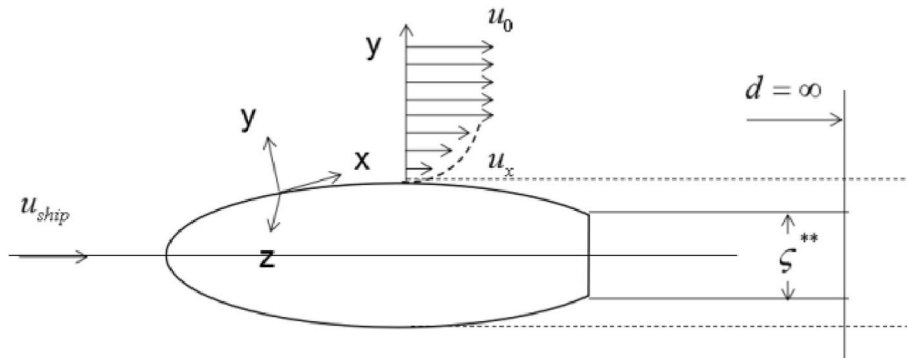


Fig. 2. schematic map of turbulent boundary layer flow around a ship body. ζ^{**} is thickness of momentum deficit.

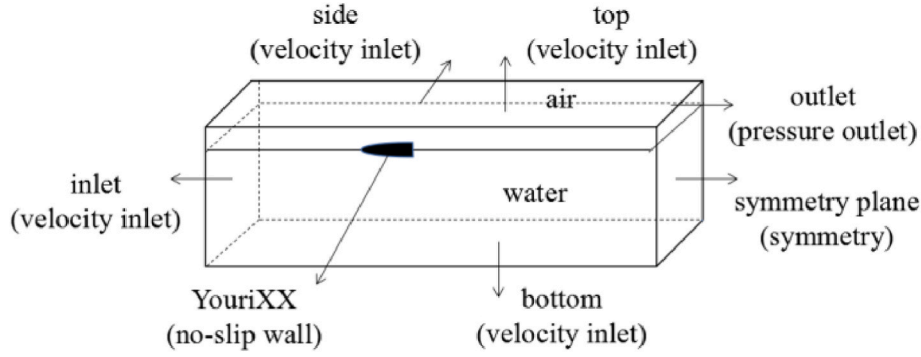


Fig. 3. an overview of the domain with the selected boundary conditions.

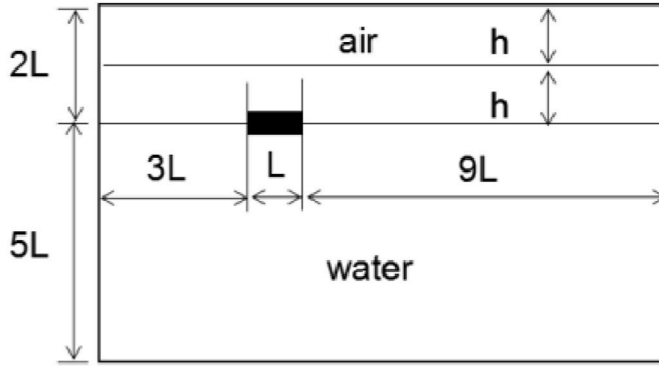


Fig. 4. the positions of the boundaries. h represents height of the air and submerged depth of the ship hull. L is length of the ship between perpendiculars.

G_k is an added term of k due to velocity gradient,
 G_b is an added term of k due to buoyancy,
 Y_M is the contribution of turbulent pulsatile expansion,
 $C_\mu, C_{1\varepsilon}, C_{2\varepsilon}$ and $C_{3\varepsilon}$ are empirical constants,
 σ_k and σ_ε are the prandtl numbers of the turbulent kinetic energy k and dissipation rate ε , respectively,
 S_k and S_ε are user-defined source terms.

We find, however, that this model is suitable for general turbulent boundary layers, but it is difficult to reflect the real characteristics of the turbulent boundary layers when the hull surface becomes extra rough, such as fouled by heavy calcareous biofoulers, with the simulations somewhat distorting.

In an attempt to overcome this shortcoming, modification of the turbulent momentum is proposed after fully considering the possible rotating and swirling of fluid flowing over high, hard and sharp biofoulers. The calibrated terms can be expressed as equations (7) and (8), respectively:

$$\frac{\partial}{\partial x_j} \left[\alpha_k \mu_r \frac{\partial k_r}{\partial x_j} + \beta_k \mu_s \frac{\partial k_s}{\partial x_j} \right] + G_k + \rho \varepsilon \quad (7)$$

The modified transport equation of k can be obtained by substituting equation (8) for the right term of equation (5).

$$\frac{\partial}{\partial x_j} \left[\alpha_\varepsilon \mu_r \frac{\partial \varepsilon_r}{\partial x_j} + \beta_\varepsilon \mu_s \frac{\partial \varepsilon_s}{\partial x_j} \right] + \frac{C_{1\varepsilon} \varepsilon}{k} G_k - C_{2\varepsilon} \rho \frac{\varepsilon^2}{k} \quad (8)$$

Likewise, the other one for ε can be given by substituting equation (8) for the right term of equation (6).

Where μ_r is a rotating-related variable mainly determined by height and hardness of the biofoulers and μ_s is a swirling-related variable primarily determined by shape and density of the biofoulers which are

proposed by Xiong and Yuan (2016) based on empirical formulae. α_ε and α_k , β_ε and β_k all are empirical constants. These constants are obtained through either experiment or experience (Xiong and Yuan, 2016) and used by others (Yao et al., 2017). Computed results with these constants are very close to measured, indicating that they are suitable for the models.

The computational domains are discretized and solved using a finite volume method while the momentum equations are discretized utilizing the second-order upwind convection scheme and a first-order temporal discretization method (Song et al., 2019). The time step value of the simulated ship hull is set to $\sim 0.0005 L_{BP}/U$, which is almost ten times smaller than the minimum value recommended in the ITTC (2011), to guarantee a robust result for such a complicated simulation.

The roughness function model established using the experimental data from Schultz and Flack (2007) is employed in the wall-function of the CFD software package STAR-CCM+ (Demirel et al., 2017).

2.5. Components of ship resistance coefficient

For a general cargo ship, the total drag is primarily composed of wavemaking resistance (induced by both pressure and viscosity) and frictional resistance. For a heavily-fouled ship, however, the viscous resistance and form resistance should be taken into consideration. Therefore, the total drag shows as:

$$R_T = R_f + R_w + R_v + R_{fm} \quad (9)$$

For a slow-speed ship ($Fr < 0.18$), the frictional resistance usually accounts for about 70–80% of the total resistance (Xiong and Yuan, 2016) while the viscous resistance for about 10% (wavemaking resistance is very small); but for a high-speed ship ($Fr > 0.30$), the frictional resistance normally accounts for about 40–50% of the total resistance while the wave resistance for about 50%. Most merchant ships are slow- or moderate-speed ships. However, for a heavily-fouled cargo ship despite its slow speed, the above proposition is no longer applicable and should be re-defined. This can be seen in the model experiment in the next section.

Accordingly, the total drag coefficient is shown in (10).

$$C_T = C_f + C_w + C_v + C_{fm} \quad (10)$$

Where the total drag coefficient C_T involves frictional resistance coefficient C_f , wavemaking resistance coefficient C_w , viscous resistance coefficient C_v and form resistance coefficient C_{fm} .

3. Results

Marine biofoulers may vary in size (height), shape, hardness and density. Each of these characteristics can have its own unique impacts on the roughness of ship hulls, in turn, the hydrodynamic performance. From the hydrodynamic perspective, however, an estimate of the ship

total drag just based on the height of the marine biofoulers on the hulls is insufficient, and can not accurately reflect the roughness of ship hulls. Therefore, a coefficient allowance determined by other factors, such as hardness, shape, density of the biofoulers, etc., can greatly add the accuracy of resistance computation. Following analysis of ship resistance is done based on real specifications of the vessel “YOURIXX” (Table 1) using FLUENT software. A comparison is also provided before and after modifications of the prescribed standard $k-\epsilon$ model in FLUENT.

3.1. Configuration of geometrical models

In order to numerically simulate the ship resistance under the mild, moderate and heavy fouling conditions, all factors which contribute to the coefficient allowance are suggested to be parameterized following the scheme of Yao et al., (2017) and converted into computational models. The modified standard $k-\epsilon$ model is chosen as the physical model. However, the original standard $k-\epsilon$ model is still remained for comparison of the outputs. An expansion of the FLUENT code using C language based on UDF is dynamically added to the existing FLUENT codes by re-editing programme in terms of the modified model. Details of grid numbers, flow fields, computational domain, start and end positions, and other ambient environment, are available in ref (Demirel et al., 2017).

Based on the three biofouling conditions, we categorize the shape of the biofoulers into 11 grades from 0 to 10. Grade 0 represents hydrodynamic smooth, grades 1–3 mild fouling, grades 4–7 moderate fouling and grades 8–10 heavy fouling. Similarly, other parameters, such as height, hardness, density of the biofoulers can be arranged in the same way. Fig. 5 shows the three types of the geometrical models.

3.2. Mesh construction

The accuracy of the results largely relies on the numbers and arrangements of the grids which are created automatically by a cell generator. The grids are usually not distributed evenly, rather, inhomogeneously, with finer and more grids in some vital areas for capturing immediate changes of related physical parameters, such as the area immediately around the bulb bow, rudder, propeller, intake of cooling water, etc. Additional mesh refinements should also be applied to layers close to ship hulls (Fig. 6).

The total resistance coefficients for each mesh configuration are computed at a design speed of 13.5 knots and are given in Table 2 for full-scale simulation of the ship hull.

The vessel model tests are performed on a 1:50 scale model after the grids are imported into FLUENT because the current CFD is difficult to simulate the ships with the real full-scale size (Yao et al., 2017).

3.3. Evolution of ship resistance coefficient with biofouling

The most noticeable feature of the modified standard $k-\epsilon$ model is that it can estimate the contribution of each single parameter to the coefficient allowance involved in the biofouling (Table 3). Table 3 shows that when the fouling condition is relatively light, the residuary

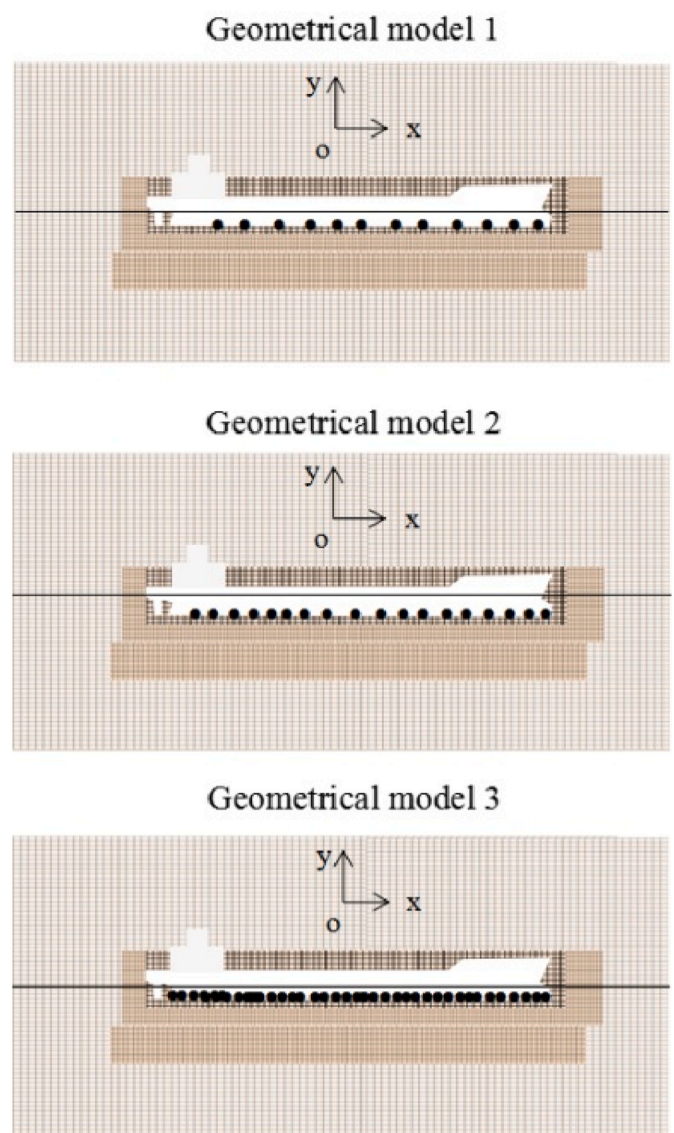


Fig. 5. three types of geometrical models for fouled vessel “YOURIXX”. a, mild fouling (top). b, moderate fouling (middle). c, heavy fouling (bottom).

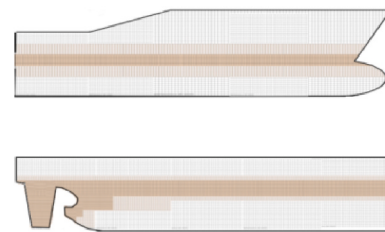


Fig. 6. mesh models. a, mesh for turbulent boundary layers (top). b, mesh for some critical areas of the fouled bottom and stern (bottom).

Table 1

Main ship specifications for “YOURIXX”.

Built Year	1996
Waterline Length LWL	278.82 m
Displacement DIS	92951.26 m ³
Breadth B	49.96 m
Draft	19.8 m
Wetted Surface Area SW	13752.17 m ²
Block Coefficient C _b	0.852
Service Speed u	13.5 Kts
Shaft Power P at Design Speed	8958.26 KW
Propulsive Coefficient	0.640

Table 2

C_T results at different mesh configurations for the full-scale ship hull at 13.5 knots.

mesh configuration	total No. of cells	C _T
coarse	4.86×10^6	0.00683
medium	6.02×10^6	0.00704
fine	8.06×10^6	0.00732

Table 3
contributions of each single factor to resistance coefficient for YOURIXX (ΔC_T)(U = 13.5 kts).

	mild fouling				moderate fouling				heavy fouling			
	C_{fm}	C_v	C_f	C_w	C_{fm}	C_v	C_f	C_w	C_{fm}	C_v	C_f	C_w
height	3.6E-04	1.2E-04	4.8E-04	1.4E-03	4.5E-04	2.5E-04	7.9E-04	9.9E-04	6.3E-04	4.2E-04	1.1E-03	1.7E-04
shape	2.1E-04	7.0E-05	2.8E-04	8.3E-04	2.6E-04	1.5E-04	4.6E-04	5.8E-04	3.6E-04	2.4E-04	6.1E-04	2.5E-04
hardness	1.3E-04	4.0E-05	1.7E-04	5.2E-04	1.6E-04	2.0E-05	2.9E-04	3.6E-04	2.3E-04	3.0E-05	3.8E-04	6.0E-05
density	2.8E-04	9.0E-05	3.7E-04	1.1E-03	3.5E-04	1.9E-04	6.2E-04	7.8E-04	4.8E-04	3.2E-04	8.1E-04	1.2E-04
C_b and bulb bow shape	2.0E-05	1.0E-05	3.0E-05	8.0E-05	2.0E-05	1.0E-05	4.0E-05	6.0E-05	4.0E-05	2.0E-05	6.0E-05	1.2E-03
sub total ΔC_T	9.9E-04	3.3E-04	1.3E-03	3.97E-03	1.24E-03	6.90E-04	2.21E-03	2.77E-03	1.74E-03	1.16E-03	2.91E-03	1.45E-03
G. total ΔC_T	6.62E-03				6.91E-03				7.26E-03			

resistance coefficient C_w dominates, accounting for 60% of the total resistance coefficient C_T , the frictional resistance coefficient C_f for about 20%, the form resistance coefficient C_{fm} for about 15% and the viscous resistance coefficient C_v for about 5%. However, as the vessel is fouled by the heavy fouling condition, above proportion changes dramatically, with the C_w accounting for only about 20%, C_f for about 40%, C_{fm} for about 24% and C_v for about 16%, indicating that the frictional resistance coefficient prevails while the shape and viscosity of the biofoulers play a considerable role due to slimes secreted by some sizable macro-organisms.

Table 3 further demonstrates that apart from height of the biofoulers, other parameters, such as shape, hardness and density of the biofoulers as well as the block coefficient C_b and the shape of the bulb bow always play a role under any conditions. In particular, the contribution of the block coefficient C_b and the shape of the bulb bow to the residuary resistance coefficient is prominent as the vessel is in a heavy fouling condition.

Based on the modified standard model, the total drag coefficient for vessel “YOURIXX” can be re-projected (Fig. 7). The modified model

illustrates an improved total drag coefficient with an increase of 15.5% compared with that obtained from the original model, in well agreement with the measured, implying that the modified model behaves in a reasonable manner in simulating the response of the ship resistance to the external forces arising from the biofouling.

The modified model offers us great insights for estimating the average speed of “YOURIXX”. Based on the relationship between the speed and the resistance:

$$R_T = C_T \frac{1}{2} \rho u^2 S \quad (11)$$

Where ρ is density of fluid, u is ship speed and S is the wetted surface area. The speed under the heavy condition is estimated to be about 10.4 kts, with speed loss of 3.1 kts, quantitatively consistent with the measured 3 kts. However, the speed of “YOURIXX” is projected to be 11.9 kts before the modification is applied, with a discrepancy of ~14.4%.

3.4. Trend of ship resistance change

How the ship resistance change at various speeds in each geometrical model is of great importance for the shipping industry. On the one hand, shipowners can optimize the hull maintenance based on the trend of ship resistance change and maximize the profits; on the other hand, biofouling control is a state strategy for future green shipping and proper measures should be taken by shipowners.

In an attempt to analyze the trend of ship resistance change after modification of the prescribed model, an increment of the resistance ΔR_T is proposed by running the model in each geometrical model to observe the development of the resistance in the context of various speeds.

$$\Delta R_T \% = (R_{am} - R_{bm}) / R_{am} \quad (12)$$

Where $\Delta R_T \%$ is the increment percentage of the total resistance, R_{am} is the total resistance after the model is modified and R_{bm} is the total resistance before the model is modified.

Fig. 8 reveals a growing of ship resistance increment under various fouling conditions after the prescribed model is modified but the increment is not very apparent in the mild and moderate fouling conditions except for an abrupt increase at speed of 21–23 kts. However, for the heavy fouling condition, a quick growing of ship resistance increment starts at speed of 6–7 kts, peaking at 13–14 kts for an increment of 15–16% for most slow-speed merchant vessels (excluding high-speed vessels).

Model experiments show that wave patterns remain little change even a little bit reduced in the light fouling condition after the model is modified (Fig. 9a,c) but the wave patterns rise, particularly in forward locations in the context of heavy fouling condition (Fig. 9b and d), presumably induced by the extra roughness on the forward parts of ship hulls. The wave pattern seems to be lower in the aft parts of the hulls (Fig. 9d) possibly because of the strong discharge flow, lowering the pressure around the aft parts of the hulls.

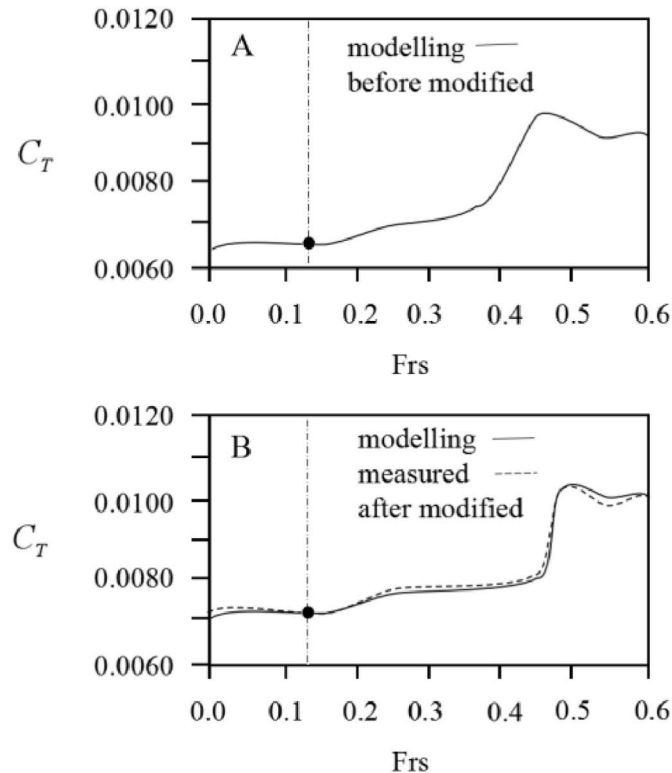


Fig. 7. measured and modeled total resistance coefficient for vessel “YOURIXX” before and after modification of the standard $k - \varepsilon$ model. The vertical dashed lines represent the Froude number at the service speed of YOURIXX. The black points denote the total resistance coefficients both measured and modeled before (A) and after (B) modification.

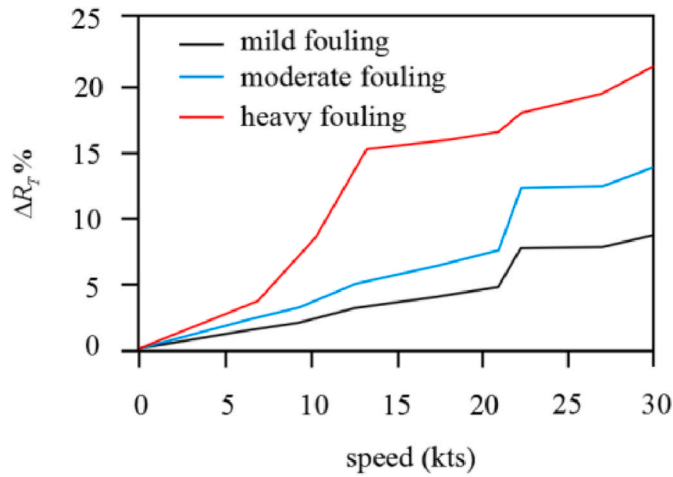


Fig. 8. general trends of ship resistance change in each fouling condition after the model is modified.

3.5. Velocity field and turbulent kinetic energy field distributions

Velocity and turbulent kinetic energy fields in hydraulic smooth and heavy fouling conditions are given in Fig. 10. As can be seen from Fig. 10, the velocity value reduces with the growth of roughness on ship hull surface whereas the turbulent kinetic energy gains with hull surface roughness. This is consistent with measured that ship hull roughness results in a stronger turbulence, which suggests that the turbulent stress and wall shear stress both increase, eventually leading to a decrease of the velocity in the turbulent boundary layers.

3.6. Validation of the results

All simulated results for the vessel “YOURIXX” have been validated against the data in the archives provided by the class and shipowner,

showing that the results obtained through the modified model is more close to the measured, particularly in the heavy fouling condition. Apart from the vessel “YOURIXX”, other ships ever being studied in previous literature (Chen and Ye, 2008; Chen et al., 2015) have also been re-analyzed by us. The methodology used by above authors is very similar to that adopted by Demirel et al. (2017). For example, the modeled $\Delta R_f/R_i$ (38–46%) in the heavy fouling condition have been compared with measured (35–42%) in the archives for the prescribed ships (Chen and Ye, 2008) and they are very close to each other, indicating that the simulated results are reliable. Another example is that under the moderate fouling condition, the modeled $\Delta R_f/R_i$ is 20–28% while it is 35–47% under the heavy fouling condition, quantitatively consistent with measured for the prescribed ships in the archives, suggesting that the model behaves in a reasonable manner. Model performance are also carried out under the various fouling conditions by modifying the time-step size from $\sim 0.005\text{LBP}/U$ $\sim 0.0005\text{LBP}/U$, and find that the error rate of R_i decreases from 0.168 to 0.022, suggesting relative importance of the time-step value. The model experiments have also shown that the modified model is suitable for any type of ships with length less than 400m but an error rate is expanded as the length of ships is less than 50m, implying a room for further improvement.

3.7. Sensitivity experiments

In an effort to examine how the fineness and density of the grids affect the simulated results, a sensitivity is tested after performance of the model experiments to test the response of the resistance to the grid fineness and density by using different fineness and density. Taking the geometrical model 3 as an example, the ship speed is set to 13.5 kts, with other conditions, such as boundary layers, ambient temperature, salinity, viscosity, pressure, etc., remaining the same. The simulated results indicate that the error rate of resistance grows with a reduce of the grid fineness and density (Table 4), implying a significant influence of the fineness and density of the grid on the final results.

Some parallel experiments are also conducted to examine the response of the resistance to the arrangements of the grids. For example,

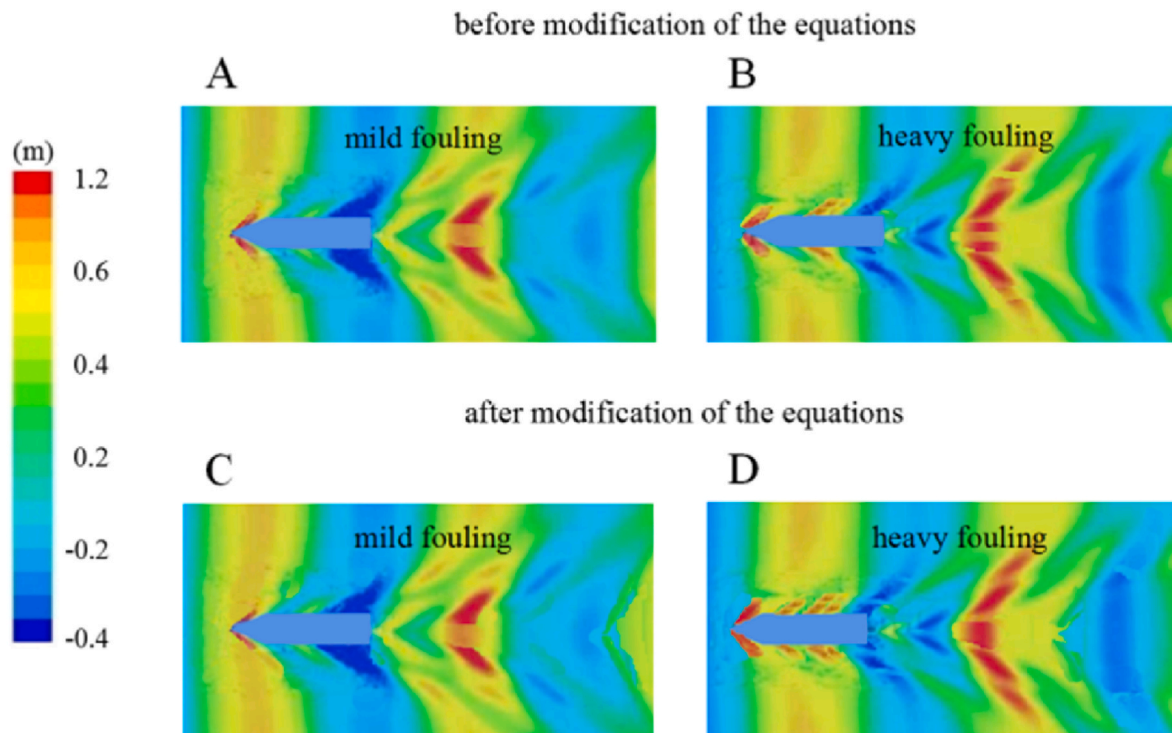


Fig. 9. comparison of wave patterns of modeled vessel “YOURIXX” at Froude = 0.12 before and after the model is modified.

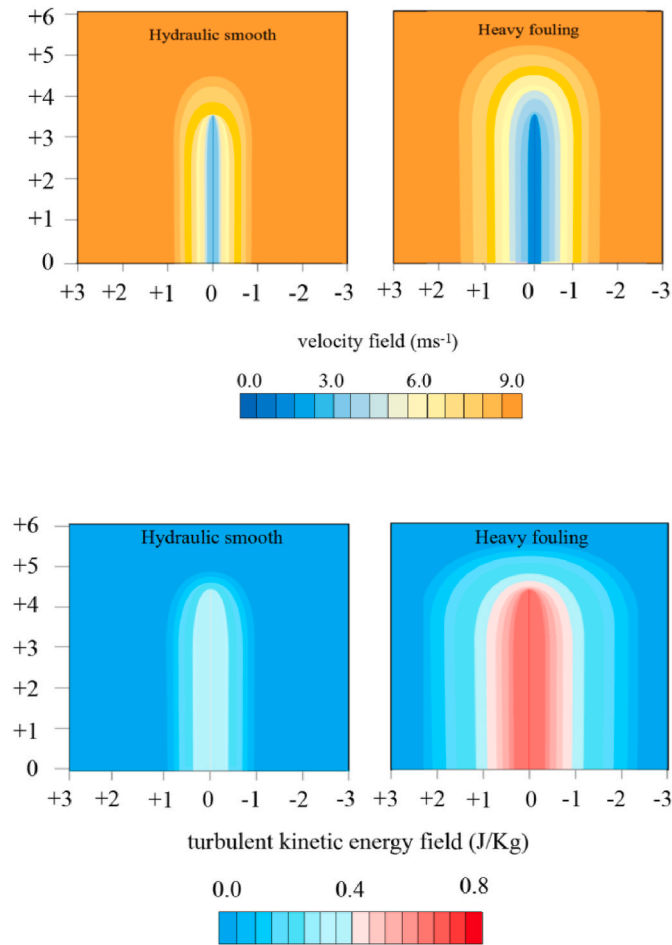


Fig. 10. Velocity field (top) and turbulent kinetic energy field distributions (bottom) at the midship of the flat plate representing the prescribed ship hull for hydrodynamic smooth and heavy fouling conditions ($V = 13.5$ knots) (grid spacing in meter).

Table 4
contrast between different mesh models.

number of grid nodes	85826	12068	202182	326842	418320
rate of resistance errors	0.56%	0.37%	0.23%	0.14%	0.06%

the model is run by gradually reducing the grid fineness and density around some critical areas, with the grids in other areas remaining unchanged. As expected, similar conclusion is yielded as above, manifesting the critical role played by the distributions of the grids.

4. Discussion

Marine biofouling widely exists in almost all underwater structures, including merchant vessels, warships, cruise ships, and other fixed objects. Previous studies on the features of biofoulers and their impact on ship resistance are largely hindered by immature understanding and less-developed numerical techniques. Development of new technology and materials for antifouling is also limited by shipping markets and policy. It is time to understand the relative importance of biofouling in reducing ship speed and causing damage to marine ecological system.

4.1. Non-negligible added resistance

Ship resistance is considered to mainly consist of the wavemaking resistance and frictional resistance. Traditionally, the frictional

resistance is computed by considering seawater viscosity superimposed on underestimated viscous resistance produced by biofoulers. In addition, other resistances, such as the form resistance caused by sharp biofoulers on ship hulls, are largely neglected. Calibrations for the discrepancy between the modeled and measured rely on the corrections of bulb bow shape coefficient and other coefficients, demonstrating the flaws of previous methods. Numerical model experiments indicate that these added resistances are considerable and play a very important role in retarding ship speeds. For example, some macrobiofoulers with a sharp shape (compared with that with a round shape) can lead to a rotating and swirling of the fluid elements and increase the thickness of the turbulent layers, thus contributing to added form and friction resistance. Another example is that harder biofoulers can remain stiff as a ship proceeds at a normal speed and produce stronger friction, form and wave making resistance than softer biofoulers. The other characteristics of the biofoulers in Table 3 can generate similar effects as above.

4.2. Effects of ship hull roughness

As described in foregoing sections, shear stress and pressure fields both are bound to be influenced by the hull roughness in the similar manner. Here the simulated pressure fields are shown in Fig. 11 as examples. The existence of heavy biofouling on the ship hull can result in not only a growing in thickness of the boundary layers through flow rotating and swirling, but also an increase of the shear stress and pressure in some critical parts, thus eventually depleting some momentum, in turn, slowing down ship average speed. Our simulations show that modified model can reflect the features of the ship resistance more accurately (compared with original model) after considering more characteristics of the biofoulers (Fig. 11).

4.3. Marine biofouling control

As stated above, the real cost of hull biofouling is actually well beyond direct costs sustained from speed loss, fuel oil penalties and extra hull maintenance. Extra carbon dioxide, sulfide and nitrogen oxides with higher-than-usual concentration are released to air (IMO, 2014), giving rise to not only greenhouse effects but also serious pollution of atmospheric environment (IMO, 2011). It is suggested in the academic circles that effectively combating shipping emissions constitute part of global climate governance (Anderson and Bows, 2012).

At present, there is no perfect method to completely prevent the hull surface from marine biofouling. Widely-used antifouling paints which contain multiple synthetic biocides or organometallic compounds (Yebara

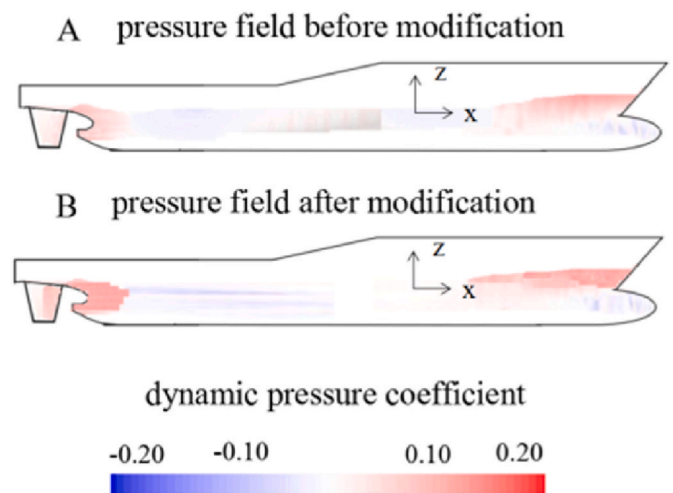


Fig. 11. simulated pressure field before modification (A) and after modification (B) for heavy fouling condition ($V = 13.5$ knots).

et al., 2004; Thomas and Brooks, 2010) are facing a global ban. Therefore the shipping industry calls for new methods that are not harmful to marine aquatic species and ocean environment while effectively protecting ship hulls. It is foreseeable that marine biofouling control has a long way to go.

5. Conclusion

In summary, marine macro-biofouler-generated hull surface roughness is different from textbook-described roughness. The hydrodynamic resistance arising from the former can not be precisely solved by traditional fluid mechanics. Therefore, some conclusions can be drawn:

- 1) Ship hull roughness involving large number of marine macro-foulers is more complex than that described in textbooks.
- 2) Viscous resistance, frictional resistance and form resistance are unrelated to homogeneity, instead, associated with density (coverage), height, types and shapes of the aquatic species attached on the hulls.
- 3) Different types of the marine biofoulers on ship hulls have different impacts on the total resistance of ships.
- 4) Conventional methods in computing the viscous resistance induced by seawater viscosity are not applicable to that caused by the macro-biofoulers.
- 5) Ship total resistance yielded by the modified model after considering the possible rotating and swirling of fluid flowing over the rough surface has an improvement of ~15%, more close to measured.
- 6) Strength of adhesion is associated with length of idle period which can be partially controlled by seafarers (Most of the biofouling take place under static or stationary conditions) (Dürr and Thomason, 2009). Longer the idle period, stronger the adhesion (Brock et al., 1999). Therefore, shortening each single idle period is feasible by intermittently heaving up anchors and taking a temporary tour around the anchorage at full speed if the anchoring is more than 2 weeks, especially in tropical regions.

Future work will focus on the experiment in estimating the relationship between ship total drags and hull surface roughness, by taking a towing tank experiment or by using steel plates with different biofoulers, immersed in a circulating water tank (water depth is up to 15 m). The anomalies of the hull surface shear stresses can be directly measured and compared with that obtained from the numerical model experiments.

CRediT authorship contribution statement

Youjia Zou: Writing – original draft, Methodology. **Xixi Zhou:** Data curation. **Lixiong Chen:** Validation. All authors equally contributed to this work and should be co-first author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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